

INSTANT RAMEN AMAZON REVIEW ANALYSIS

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Introduction

Instant ramen is globally recognized as a food that occupies a unique position in American consumerist culture. It has quickly become an easily accessible staple across diverse communities: race, income, and geography. In particular, it has held a grasps among college students and low-income households due to its low costs, convenience, and long shelf life. However, despite its popularity, ramen is often associated with health concerns such as high sodium content and low nutrition-to-calorie value. Ramen also holds strong personal and cultural expectations around taste and quality. The differences in attitude towards the product make it an expectional target to textually analyze.

This project uses sentiment analysis and topic modeling on Amazon ramen reviews to examine how consumers express attitudes toward instant ramen products in the United States.

Data

Data was taken from Kaggle. Each entry is one of 500,000 fine food Amazon reviews dating from Oct 1999 to Oct 2012. The study contains 256,059 users whose eligibility requires writing over 50 reviews for any kinds of products on Amazon. After narrowing the set to just ramen by filtering reviews with words "ramen", "instant noodle(s)" and "cup noodle(s)" we were left with 1241 entries of data and 545 unique products what were reviewed.

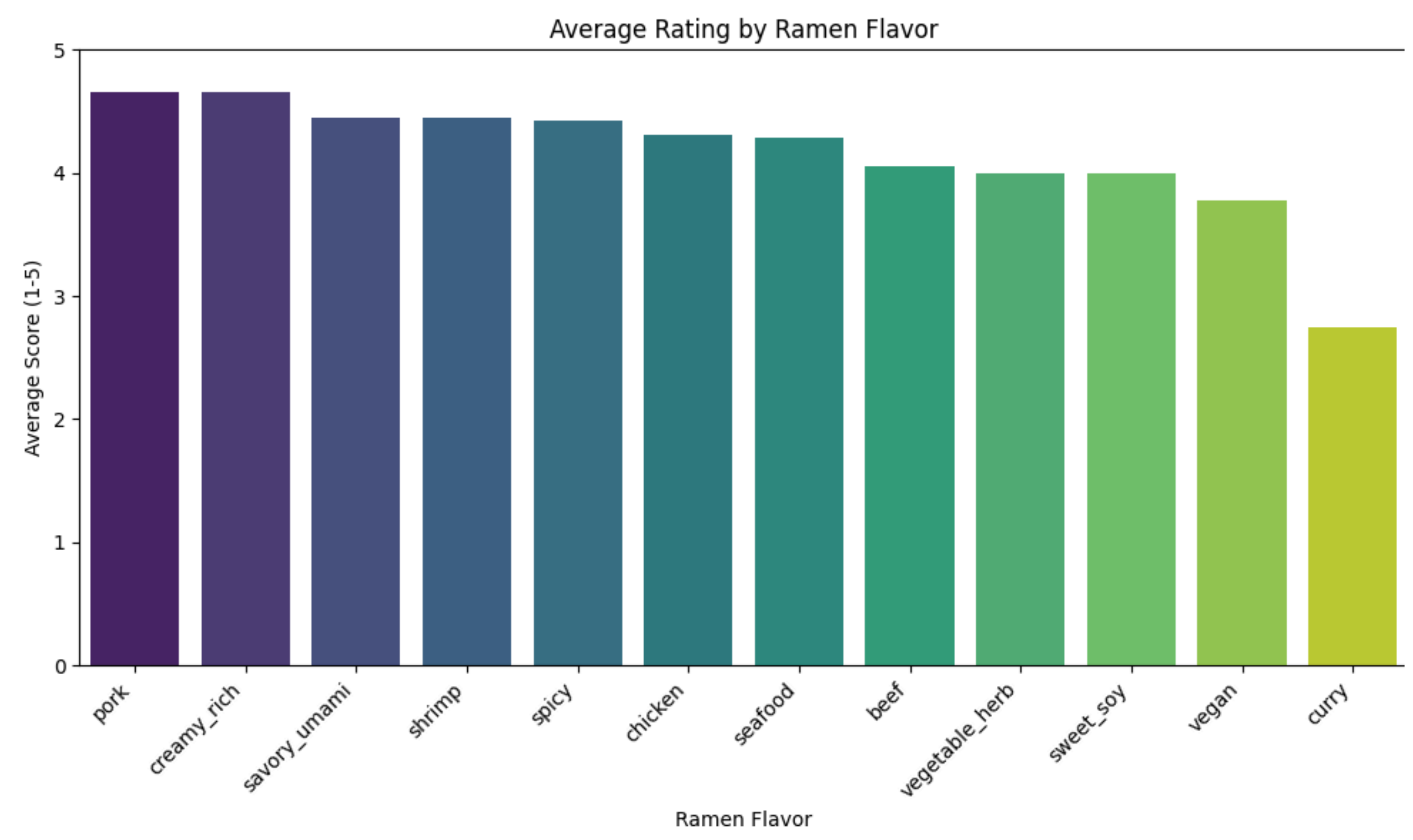
The dataset's fields are as follows: Review ID, ProductID, UserID, ProfileName, Helpfulness Numerator, HelpfulnessDenominator, Score(Star Rating), Time, Summary, and the Review Text.

Methodology

- **VADER sentiment analysis** to measure the emotional tone of ramen reviews and compare it against the Amazon star ratings. VADER was chosen because it is specifically optimized for short, informal, consumer-style text.
- **Topic Modeling (LDA)** to automatically extract recurring themes (e.g., "Texture," "Spice Level," "Value") from thousands of reviews.
- **Network Analysis:** to construct flavor-occurrence network to visualize how proteins and variations are interconnected
- **Descriptive Analysis:** Analyzed differences in average scores across 12 flavors to identify leaders and outliers
- **Altair Implementation:** Built interactive visualizations to allow users to filter by protein or score, providing a dynamic look at the ramen landscape.
- **Machine learning model using SVM** to predict the ratings of a review.

Findings

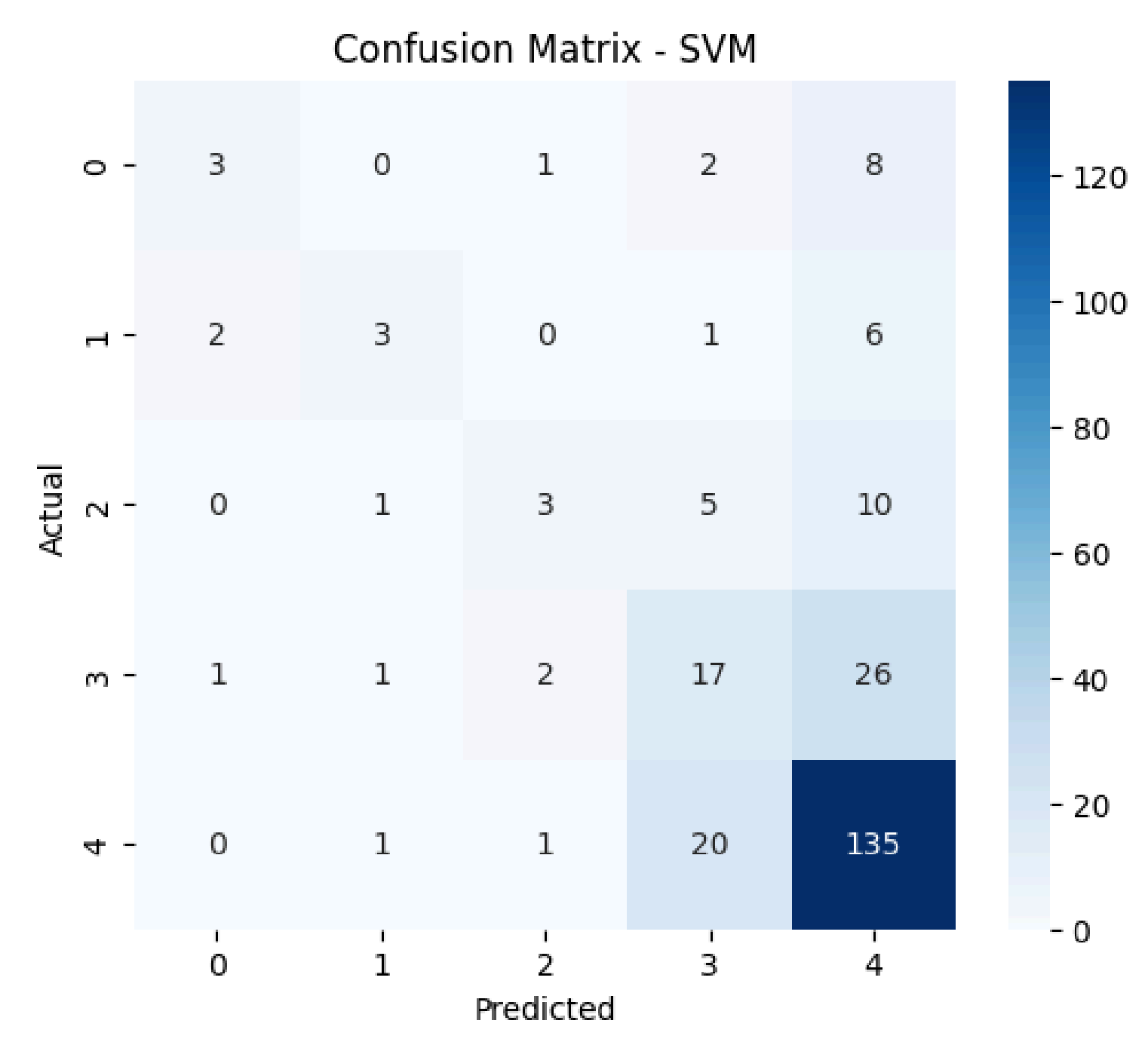
- **Descriptive Analysis**
 - **Pork** and **Creamy Rich** flavors are the highest rated (4.6/5).
 - **Curry** is a significant negative outlier with a rating of 2.7.
 - Most flavors cluster between 4.0 and 4.4, showing a high baseline for quality.
- **Topic Modeling (Themes)**
 - **Topic 1 (Rich/Meaty):** Dominates the high-scoring group (Pork, Beef).
 - **Topic 2 (Light/Plant):** Includes Vegan and Veggie options with moderate scores.
 - **Topic 3 (Seafood):** Distinct cluster featuring Shrimp and Seafood profiles.
- **Network Analysis**
 - **Pork** serves as the central "hub" node for most savory pairings.
 - Strongest connections exist between **Spicy** and **Meat-based** profiles.
 - **Curry** is visually isolated, reflecting its unique (and less popular) profile.
- **Altair Visualization**
 - Used an interactive **ranked bar chart** with a **Viridis color scale**.
 - Tooltips allow for quick data inspection of specific flavor ratings.



Machine Learning Model: SVM Classifier

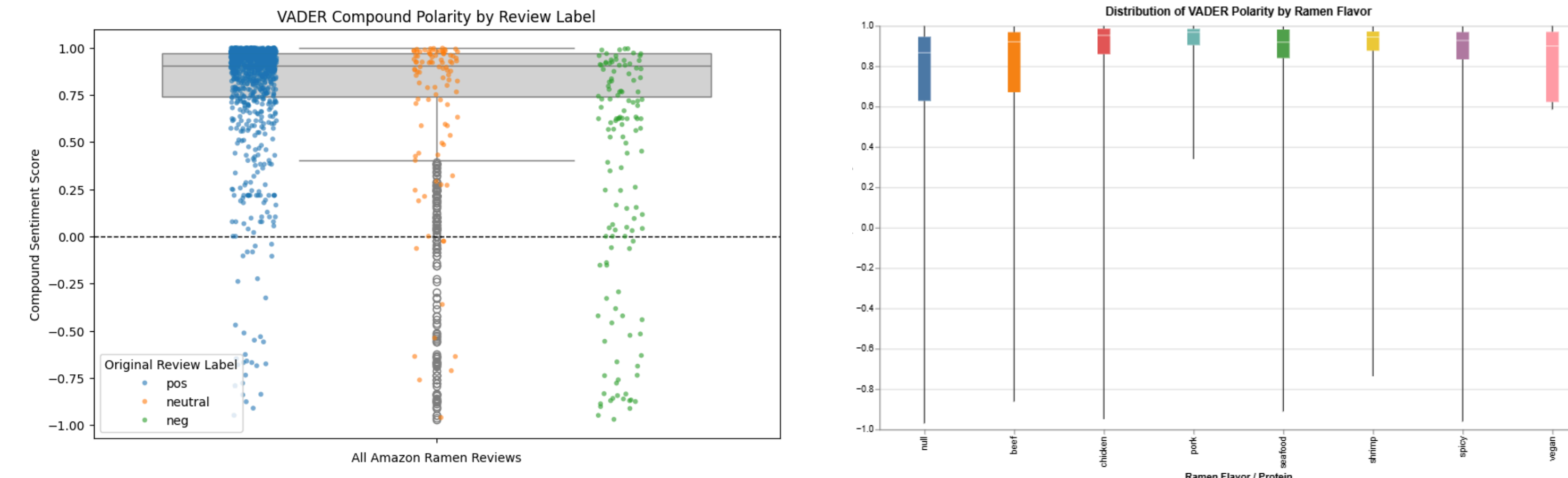
Out of 5 classifiers, the Support Vector Machine (SVM) achieved the highest accuracy ($\approx 66\%$) and the strongest overall performance.

- Accuracy: 0.65
- Weighted F1-Score: 0.62
- Best performance on 5-star reviews (F1 = 0.79)
- Moderate performance on 4-star reviews (F1 = 0.37)
- Lower performance on 1-3 stars due to class imbalance



Sentiment Analysis

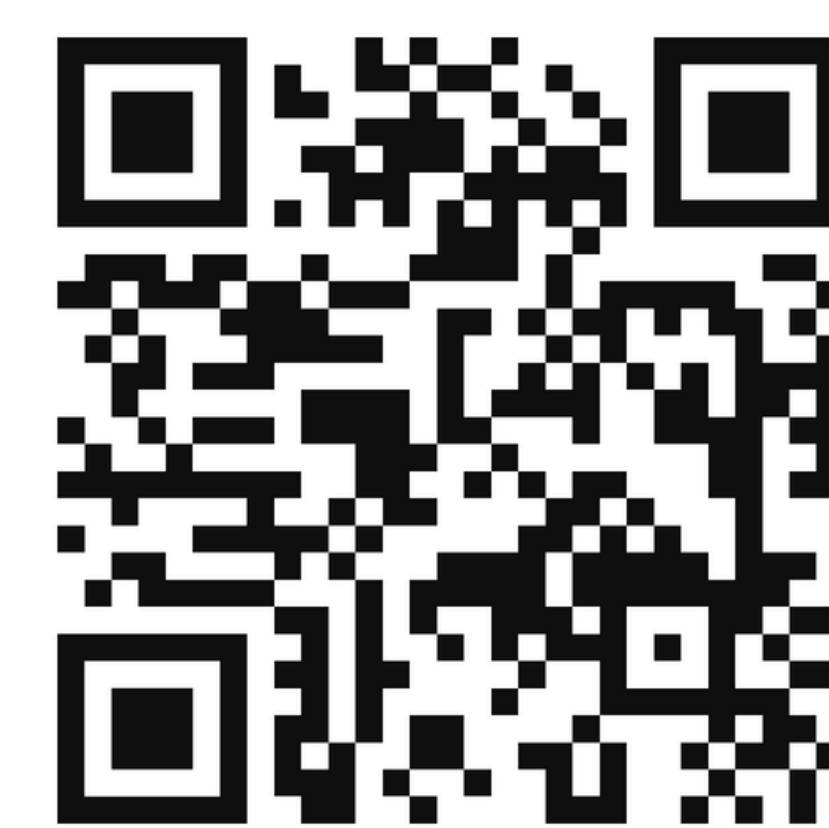
- **Strong positive sentiment:** Mean VADER polarity is 0.75, with a median of 0.91.
- **Emotional extremes:** Sentiment scores range from approximately, -0.99 to +0.99.
- **Clustered positivity:** A standard deviation of 0.40 suggests moderate variation, with most reviews concentrated as positive
- **Model-human agreement:** VADER sentiment classification matches rating-derived labels with 82.43% accuracy showing strong relationships between text sentiment and star ratings.
- **Neutral language challenge:** Neutral reviews are harder for VADER to classify due to less emotionally explicit language.
- **Vegan reception:** Vegan ramen reviews remain overwhelmingly positive (median ≈ 0.93), despite fewer product options.



Conclusions

Using consumer review text combined with rule-based sentiment analysis, we transformed unstructured Amazon ramen reviews into measurable indicators of consumer perception. This approach makes it possible to examine sentiment patterns at scale and to compare attitudes across dietary categories and flavor profiles in ways that would be difficult to assess through manual reading alone. At the same time, the analysis inherits limitations from the original data and tools, including rating subjectivity, mixed sentiment within single reviews, and occasional misclassification by VADER. As a result, sentiment scores should be interpreted as indicators of language patterns rather than definitive judgments of product quality. Despite these constraints, the workflow demonstrates how text mining can reveal meaningful insights about everyday foods that are often taken for granted.

Links



GitHub Reposeotory



Original Dataset